

1 (a) scalar has magnitude/size, vector has magnitude/size and direction B1 [1]

(b) acceleration, momentum, weight B2 [2]
 (–1 for each addition or omission but stop at zero)

(c) (i) horizontally: $7.5 \cos 40^\circ / 7.5 \sin 50^\circ = 5.7(45) / 5.75$ not 5.8 N A1 [1]

(ii) vertically: $7.5 \sin 40^\circ / 7.5 \cos 50^\circ = 4.8(2)$ N A1 [1]

(d) *either* correct shaped triangle M1
 correct labelling of two forces, three arrows and two angles A1
or correct resolving: $T_2 \cos 40^\circ = T_1 \cos 50^\circ$ (B1)
 $T_1 \sin 50^\circ + T_2 \sin 40^\circ = 7.5$ (B1)
 $T_1 = 5.7(45)$ (N) A1
 $T_2 = 4.8$ (N) A1 [4]
 (allow ± 0.2 N for scale diagram)

2 (a) 1 constant velocity / speed B1 [1]